STUDYNEST | CLASS 11 PHYSICS

QUESTION PAPER 1

Chapter 11: Thermodynamics

Maximum Marks: 40 | Suggested time: 60 minutes

Instructions: Answer all questions. Show calculation steps and units. Original practice paper, not an official examination.

Section A 1-mark questions - 8 x 1 = 8 marks

1. State the zeroth law. [1]
2. Define internal energy. [1]
3. First law with W done by system? [1]
4. Meaning of adiabatic? [1]
5. Meaning of isothermal? [1]
6. Can a cyclic heat engine be 100% efficient? [1]
7. Ideal gas equation? [1]
8. Carnot efficiency? [1]

Section B 2-mark questions - 6 x 2 = 12 marks

9. At constant volume, 500 J heat added: work? [2]
10. Gas absorbs 800 J and does 300 J work: DeltaU? [2]
11. Carnot T_h=600K,T_c=300K: efficiency? [2]
12. Ideal-gas isothermal process: DeltaU? [2]
13. 1 mol with C_v=20 J/mol K and DeltaT=5 K: DeltaU? [2]
14. Isothermal versus adiabatic? [2]

Section C 3-mark questions - 4 x 3 = 12 marks

15. Gas absorbs 1000 J and does 400 J work: DeltaU? [3]
16. Carnot T_h=500K,T_c=300K: ? [3]
17. Isobaric gas expands .01 to .03 m^3 at 100 kPa. Work? [3]
18. Why no real cyclic engine converts all heat to work? [3]

Section D 4-mark questions - 2 x 4 = 8 marks

19. Explain reversible and irreversible processes with examples. [4]
20. Engine absorbs 1500 J from 600K, rejects to 300K; actual work=500J. Find Carnot limit, actual efficiency and Qout. [4]

End of question paper.

FORMULA AND EQUATION REFERENCE

Chapter-specific equations and relations

- First law with W done by the system: Q=Delta U+W; equivalently Delta U=Q-W.
- Boundary work W=integral p dV; at constant pressure W=p Delta V; at constant volume W=0.
- For an ideal gas, internal energy depends only on temperature; for isothermal change Delta U=0.
- For n moles: Delta U=n C_v Delta T; for a monatomic ideal gas C_v=(3/2)R.
- Carnot efficiency eta=1-T_c/T_h, with temperatures in kelvin; this is the maximum possible efficiency.
- Energy balance over a cycle: Q_in=W+Q_out; actual efficiency eta=W/Q_in.
- Adiabatic process: Q=0. Isothermal process: temperature constant. Cyclic process: net Delta U=0.
- Second law: no cyclic heat engine can convert all absorbed heat into work.

Use equations only when their assumptions apply; keep units consistent.

STEP-BY-STEP SOLUTIONS

Concept questions are answered directly; numerical problems show the governing equation and substitution.

Question 1

Principle / definition: Identify the physical quantity or law asked for, then apply the chapter definition.
Apply it here: Thermal equilibrium is transitive.
Final answer: Thermal equilibrium is transitive.

Question 2

Principle / definition: Identify the physical quantity or law asked for, then apply the chapter definition.
Apply it here: Microscopic energy of molecules and their interactions.
Final answer: Microscopic energy of molecules and their interactions.

Question 3

Principle / definition: Identify the physical quantity or law asked for, then apply the chapter definition.
Apply it here: Q=DeltaU+W.
Final answer: Q=DeltaU+W.

Question 4

Principle / definition: Identify the physical quantity or law asked for, then apply the chapter definition.
Apply it here: Q=0.
Final answer: Q=0.

Question 5

Principle / definition: Identify the physical quantity or law asked for, then apply the chapter definition.
Apply it here: Temperature constant.
Final answer: Temperature constant.

Question 6

Principle / definition: Identify the physical quantity or law asked for, then apply the chapter definition.
Apply it here: No.
Final answer: No.

Question 7

Principle / definition: Identify the physical quantity or law asked for, then apply the chapter definition.
Apply it here: pV=nRT.
Final answer: pV=nRT.

Question 8

Principle / definition: Identify the physical quantity or law asked for, then apply the chapter definition.
Apply it here: 1-T_c/T_h.
Final answer: 1-T_c/T_h.

Question 9

Working: At constant volume, Delta V=0. Boundary work W=integral p dV=0. This assumes no other kind of work is involved.

Final answer: Zero boundary work.

Question 10

Working: First law gives Delta U=Q-W=800-300=500 J.

Final answer: 500 J.

Question 11

Working: eta=1-T_c/T_h=1-300/600=0.5=50%.

Final answer: 50%.

Question 12

Working: For an ideal gas, internal energy depends only on temperature. Isothermal means Delta T=0, so Delta U=0.

Final answer: Zero.

Question 13

Working: Delta U=nC_vDelta T=(1 mol)(20 J mol^-1 K^-1)(5 K)=100 J.

Final answer: 100 J.

Question 14

Working: Isothermal means temperature remains constant and heat may flow to keep it so. Adiabatic means Q=0; temperature may change as work is done.

Final answer: Isothermal: T constant; adiabatic: Q=0.

Question 15

Working: Delta U=Q-W=1000-400=600 J.

Final answer: 600 J.

Question 16

Working: eta=1-T_c/T_h=1-300/500=0.4=40%.

Final answer: 40%.

Question 17

Working: For constant pressure, W=p Delta V. Delta V=0.03-0.01=0.02 m^3; p=100 kPa=100,000 Pa. W=100,000(0.02)=2000 J.

Final answer: W=pDeltaV=2000 J.

Question 18

Working: The second law requires some heat rejection or another entropy increase for a cyclic engine. Therefore a real cyclic engine cannot convert all absorbed heat into work.

Final answer: Second law requires rejected heat or other entropy increase.

Question 19

Working: A reversible process is an ideal quasi-static process with no dissipative effects and can be reversed by an infinitesimal change. Examples: ideal slow frictionless compression. Irreversible processes involve friction, turbulence or heat flow across a finite temperature difference; real expansion with friction is an example.

Final answer: Reversible: ideal quasi-static frictionless; irreversible: friction or finite-temperature heat flow.

Question 20

Working: Carnot limit eta=1-300/600=0.50=50%. Actual efficiency=work/heat input=500/1500=0.333=33.3%. Energy balance Q_in=W+Q_out, so Q_out=1500-500=1000 J.

Final answer: Carnot limit 50%; actual efficiency 33.3%; Qout=1000 J.